Understanding and Modeling Organizational Systems

CSE 4407

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- Key Benefit: Systems principles help us understand how organizations *really* work

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- The takeaway for analysts: Always consider the wider impact of system changes

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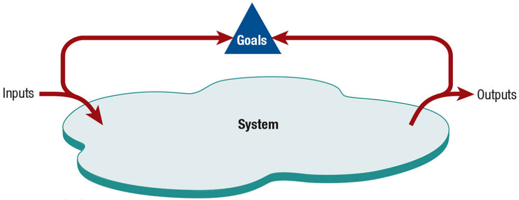
- Boundaries: Separate the system from its environment
  - Can be **permeable** (open, easy exchange) or **impermeable** (closed, restricted exchange)
  - Organizations need *some* permeability to survive (import resources, export products)

Staying on Course: Feedback and Environment

- System Control: Organizations use planning and control
- Feedback: Output is compared to goals to guide future inputs/actions
  - Example: Poor sales of red, white, and blue weight sets (output) leads to producing fewer (planning/control)



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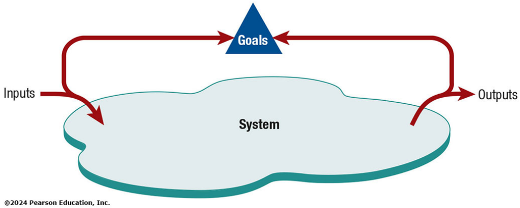

- Ideal System: Self-correcting/self-regulating
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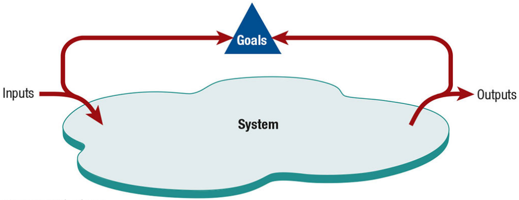

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- Environment: Everything outside the boundary. Crucial influences:
  - Community (demographics, location)
  - Economic (markets, competition)
  - Political (government regulations)
  - Legal (laws, guidelines)

Open vs. Closed and The Rise of Virtual

- Openness/Closedness: Exists on a continuum internally too
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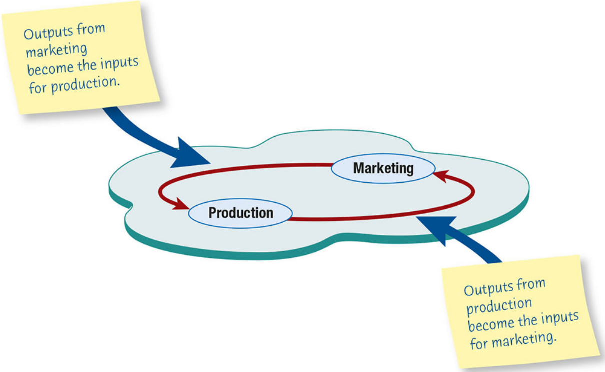
  - Many System Analysis/Design teams work virtually

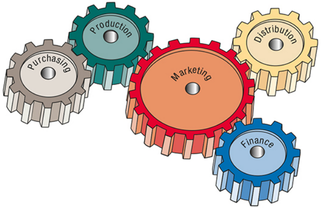
The Power of Perspective

- Systems Perspective: Allows analysts to understand the *whole* business
- Crucial Insight: Subsystems *rely* on each other. Their work is interrelated
- The Danger: Managers often see the organization through the lens of their *own* department

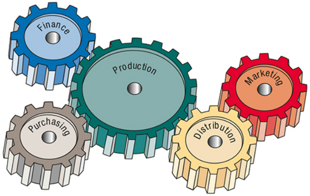


Warped Views: Managerial Perspectives

- The Problem: Managers may overemphasize their own function's importance



How a Marketing Manager May View the Organization



How a Production Manager May See the Organization

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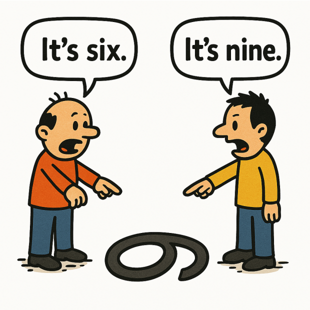
- Implication: If these managers rise to strategic levels, their biased view can lead to poor decisions for the overall organization



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- Analyst Role: Managing interfaces between ERP and existing ('legacy') systems

ERP: The Game Changer

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- Key Hurdles
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- Impact: Changes job design, required skills, and even company strategy
- Long-term View: Can make employees more effective, but requires careful planning, especially regarding user experience

Visualizing the System: Graphical Models

- Why: To understand and communicate:
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  - Relationships between system components
- Think: Blueprints and Maps for the System
- Two Key Early Models
  - Context-Level Data Flow Diagram (DFD)
  - Entity-Relationship (E-R) Model

Zooming Out: The Context-Level DFD

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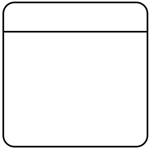

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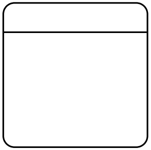

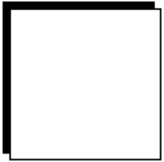

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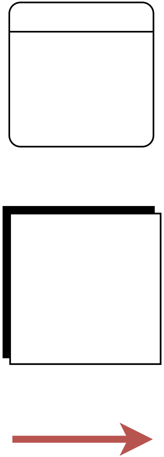
    - Data Flow: Movement of data (arrow)



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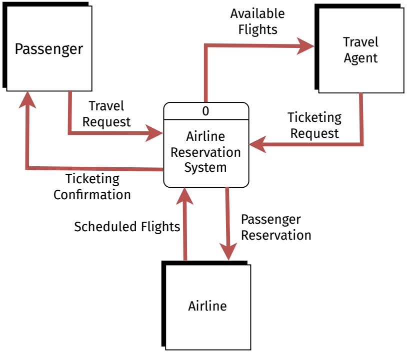

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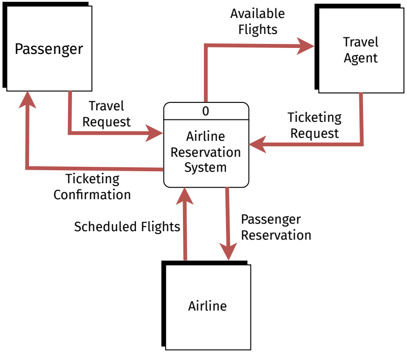

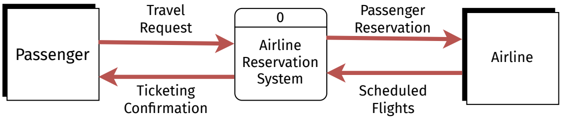

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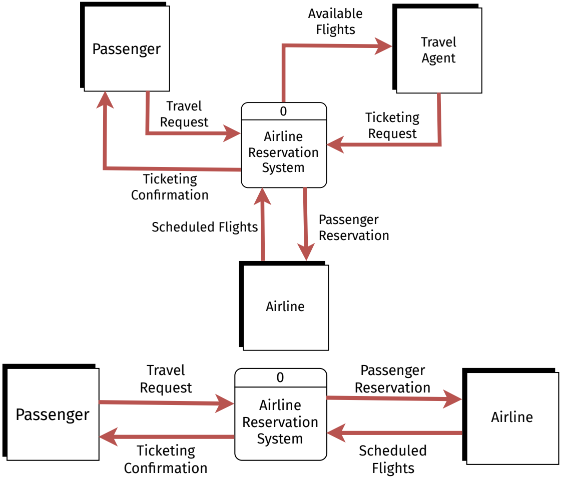
- Illustrates how context changes affect the diagram



Context-Level DFD: Scope and What It Doesn't Show

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- Use: The very first step in understanding system boundaries. A prerequisite for more detailed diagrams

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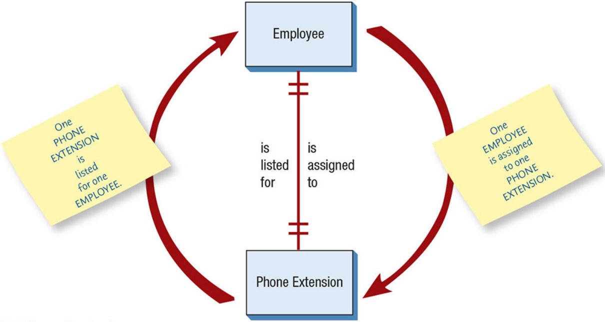
- Purpose
  - Defines the structure of the data required by the organization/system
  - Foundation of database design

E-R Basics: Reading Relationships (1:1)

- Cardinality: Specifies *how many* instances of one entity relate to *how many* instances of another



E-R Basics: Reading Relationships (1:1)

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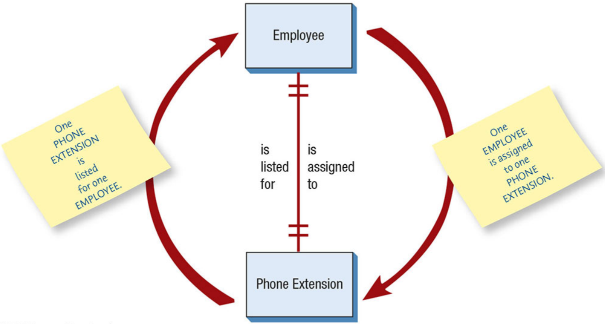
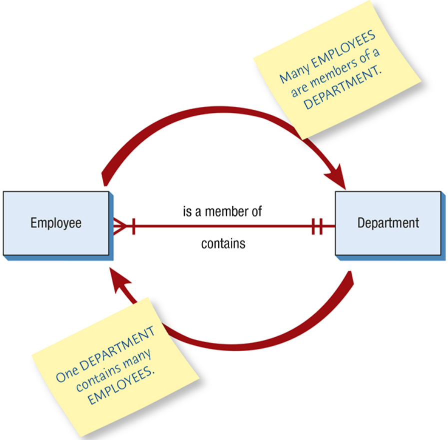
- Example: One-to-One (1:1)
    - One EMPLOYEE is assigned to exactly one PHONE EXTENSION
    - One PHONE EXTENSION is listed for exactly one EMPLOYEE
    - | | (two short parallel marks) often means ‘exactly one’



E-R Relationships: One-to-Many (1:M)

- Example: One-to-Many (1:M)
  - Many EMPLOYEEs *are members of* one DEPARTMENT (Many side)
  - One DEPARTMENT *contains* many EMPLOYEEs (One side)
- Notation
  - $>|$ (Crow's foot) often means 'many' (zero, one, or more)
  - $||$ still means 'exactly one'
- Read the relationship name *towards* the entity you're going to



E-R Notation: Cardinality and Optionality

- Common Crow's Foot Symbols
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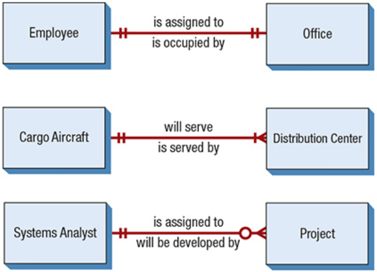

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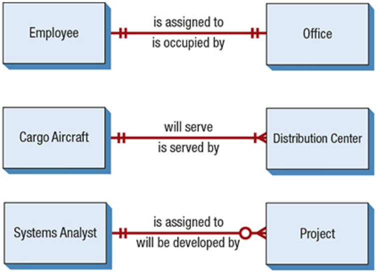

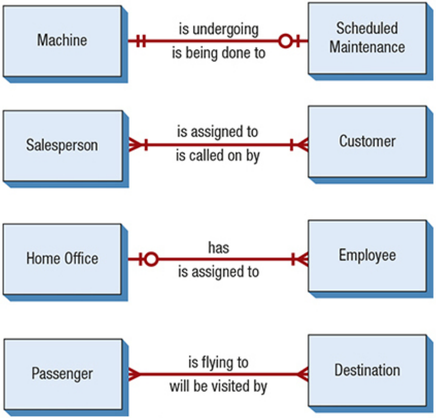

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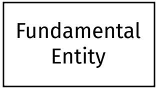
  - Example: `STUDENT`, `DEPARTMENT`, `COURSE`



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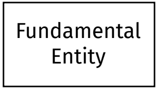
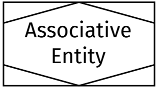
  - Links two or more other entities, often representing an event or transaction that depends on them
  - Used especially for M:M relationships
  - Example: An `REGISTRATION` links a `COURSE` and a `STUDENT`.





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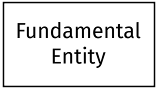

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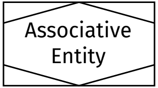
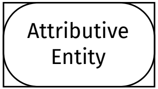


- Attributive Entity (Rectangle with curved corners)
  - Describes attributes, especially for repeating groups, of another entity
  - Dependent on a fundamental entity
  - Example: `OFFERING` might be an attributive entity holding specific `semester` and/or `year` for a `COURSE`



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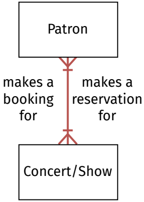

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- Evolution of Understanding
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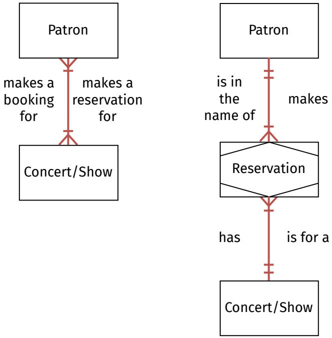

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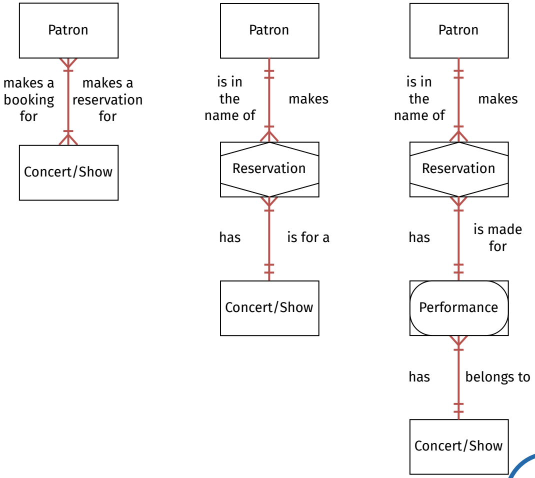

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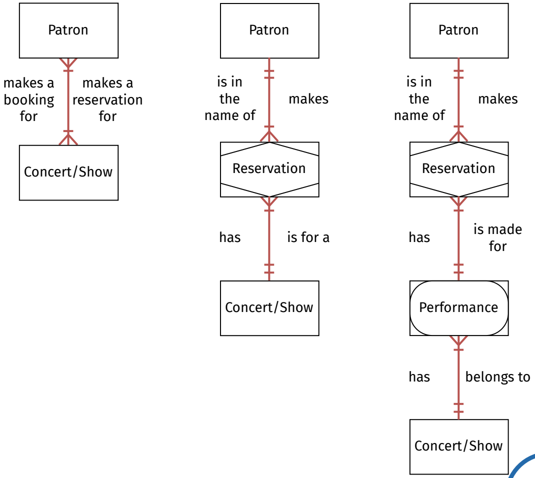
- Shows how E-R models evolve as understanding deepens
- Attributes listed for each entity. Underlined = Key/Searchable



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- Don't wait for database design! It's an analysis tool first.

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- Think: Describing user goals and how the system helps achieve them

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- Use Case: Represents a specific goal or task an actor wants to achieve with the system
  - Describes a sequence of interactions
  - Should provide something of *value* to the actor
  - Named with Strong Verb-Noun (e.g., “Enroll in Course”, “Submit Order”)
  - Symbol: Typically an oval

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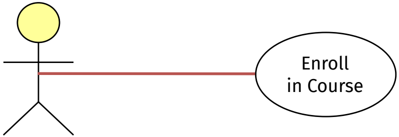
Generalizes		One actor/use case is more general version of another

Use Case Relationships Explained



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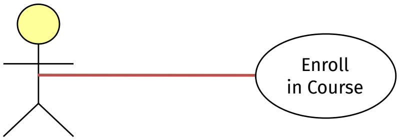

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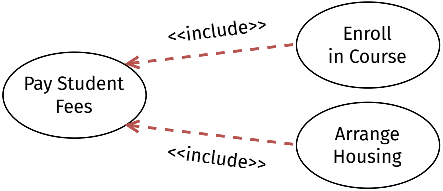

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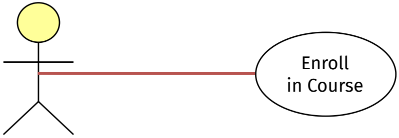

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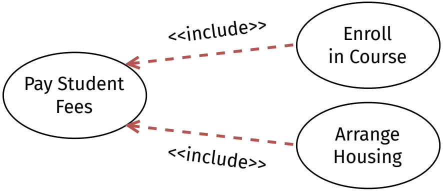

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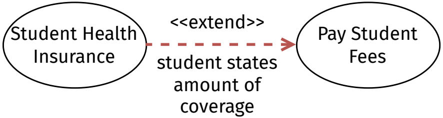

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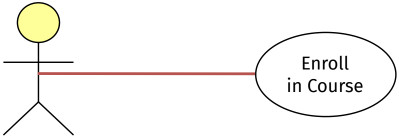

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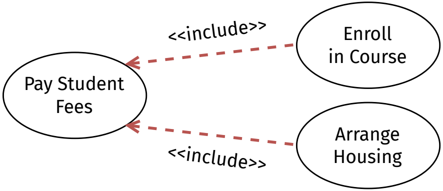

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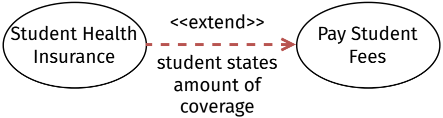

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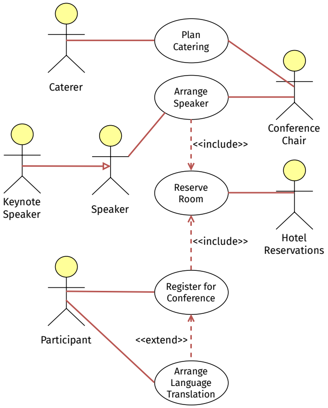
- Goal: Capture significant user interactions and goals. Keep it manageable (20-50 use cases for large systems)

Use Case Diagram Example: Conference Planning

- Scenario: System to help plan an academic conference
- Actors
  - Conference Chair
  - Participant
  - Speaker
  - Keynote Speaker
  - Hotel Reservations (Systems?)
  - Caterer



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- Purpose: Elaborate on the use case oval, providing necessary detail for design and development
- Format: Often uses a standardized organizational template

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- Choose Level Appropriately: Focus often on Blue level for user interaction understanding

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  - Header
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Demo: Use Case Scenario - Conference Registration

Use Case Scenario Example Details

- Example (Conference Registration)
  - Shows detailed steps for the main path
  - Includes conditional logic within steps (e.g., “IF multiple airports...”)
  - Can reference alternative scenarios/extensions (Flight selection, Seat selection)
  - May include looping/iterative steps
- Key: Captures the flow and necessary logic clearly for developers

Demo: Use Case Scenario - Airline Reservation

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- Understanding these levels helps us design systems that serve *everyone* effectively
- Think: Different altitudes give different views of the landscape

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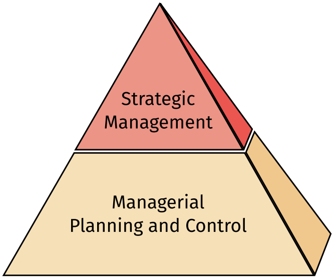

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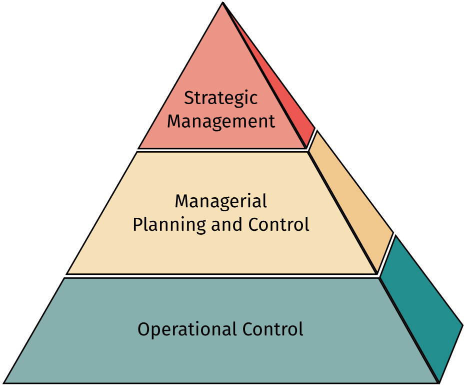

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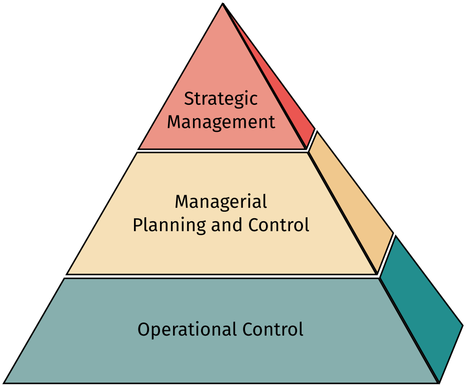

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- Moderate Need
  - Past performance (for immediate comparison)
  - Period reports (daily/weekly summaries)
- Example Systems: Transaction Processing Systems (TPS), real-time dashboards showing production status, inventory levels

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- Primary Need: Mix of internal information - current and historical
    - High need for **real-time** info (troubleshooting, control)
    - High need for **current performance** vs. **standards**
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- Moderate Need: Some external information (departmental competitors, industry trends)
- Example Systems: Management Information Systems (MIS), Decision Support Systems (DSS) providing summaries, exception reports, budget tracking, forecasting tools

Information Needs: Strategic Management

- Primary Need: External and Internal; Future-oriented
  - High dependence on **external information** (market trends, competitors, economy, regulations)
  - High need for **predictive information** and **“what-if”** scenarios (long-term strategy)
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Information Needs: Strategic Management

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  - High need for **summarized**, **periodic information** (tracking overall progress)
  - Need for **risk assessment** (including security)
- Moderate Need: Summarized internal performance (the big picture)
- Low Need: Detailed, real-time internal operational data
- Example Systems: Executive Information Systems (EIS), DSS, Business Intelligence (BI) tools providing high-level dashboards, market analysis, long-range forecasting, risk modeling

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- Analyst's Job: Understand the user's level and specific decision-making context to determine appropriate system features and information presentation

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- Opportunity: Leveraging expertise from all levels (e.g., operational users know workflows, middle managers know constraints, strategic users know goals)
- Key
  - Structure collaboration carefully
  - Value diverse expertise
  - Manage information flow

Beyond Structure: Organizational Culture

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- Why Care: Culture significantly influences
  - How people work together
  - How decisions are *really* made
  - How readily new ideas (and systems!) are accepted
- Think: The invisible forces guiding how things get done

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  - Can form around departments, teams, roles, projects, even virtual teams
  - May align with, or sometimes conflict with, the “official” culture
  - Can compete for influence (“Our way is better!”)
- Challenge: Identifying and understanding these often overlapping and dynamic subcultures

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  - Nonverbal Symbolism
    - Artifacts (awards, posters, logos)
    - Rites and ceremonies (parties, award events, retirement rituals)
    - Dress code (formal vs. casual)
    - Office layout and decoration (open plan vs. private offices, fancy vs. functional)

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- Analyst Goal: Understand subcultures to anticipate challenges and tailor implementation (e.g., targeted training)

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- Other Tools: Similar impacts from collaborative platforms (Teams, Discord, etc.), project management software, video conferencing

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- Goal: Work *with* the culture(s) where possible, mitigate conflicts, and facilitate smoother system adoption

References I

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